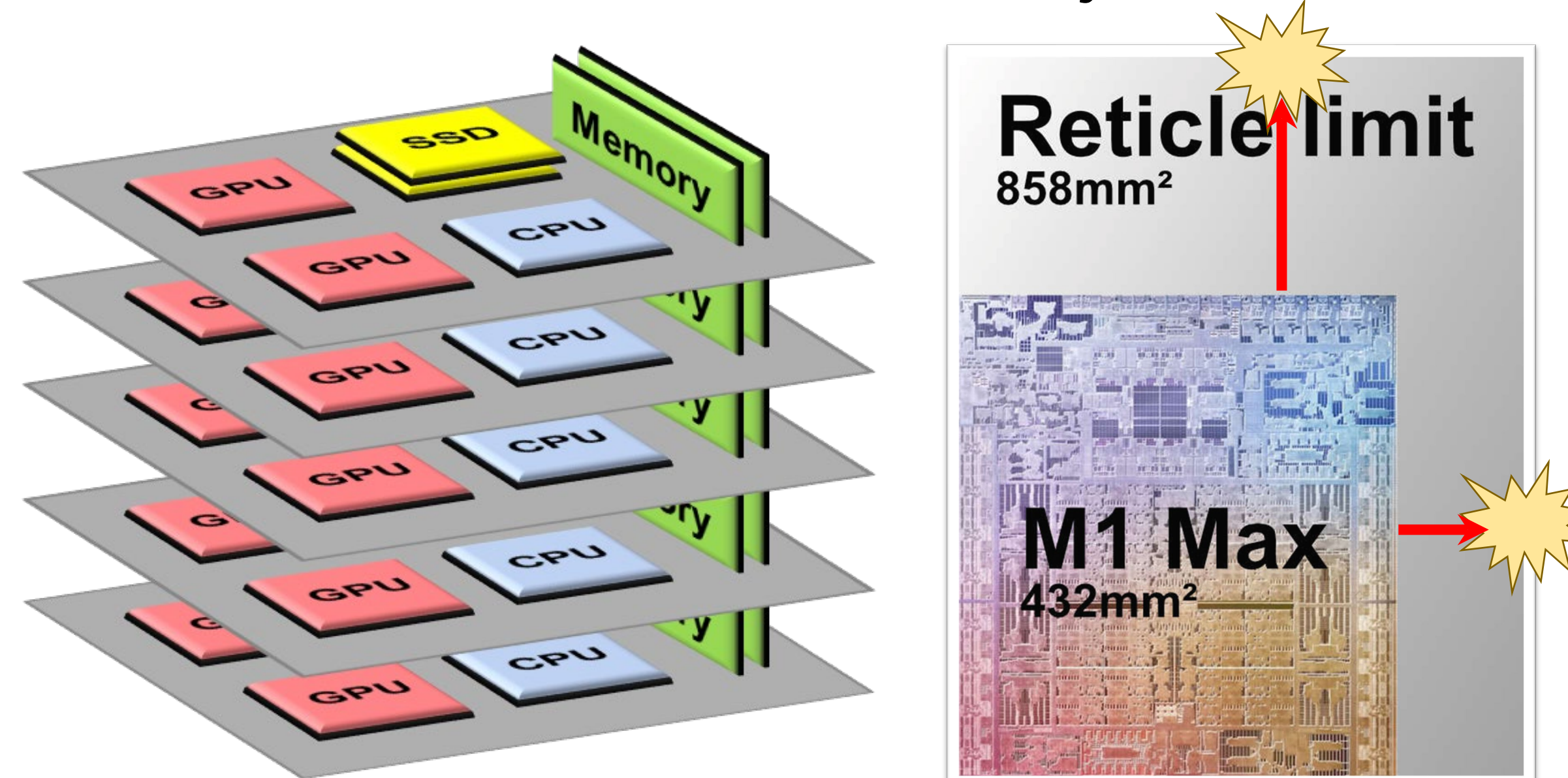




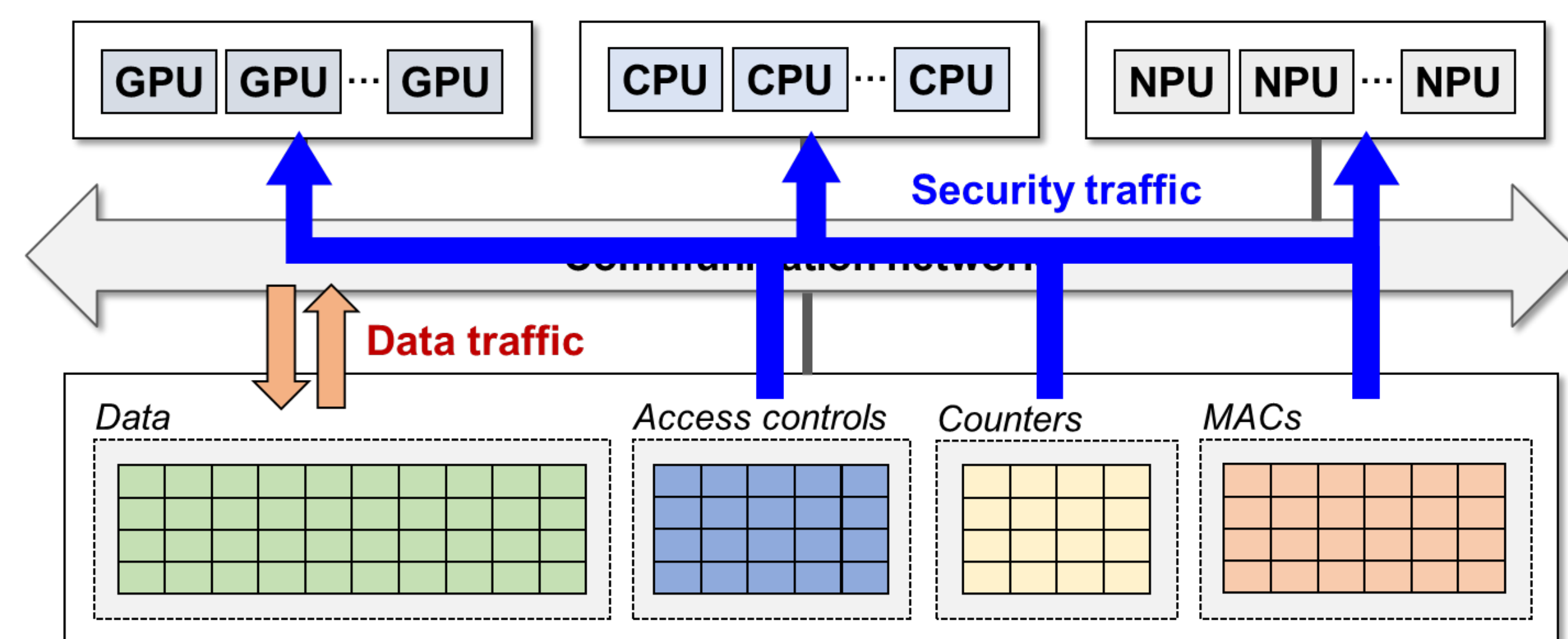
Background: Tougher Resource Expansion

- Resource expansion is becoming tougher and tougher due to form-factor limits nowadays...
 - System-level?** Few slots on motherboard & saturated pins
 - Chip-level?** Transistors on a die is reaching reticle limit
 - Reticle limit: max. area with a single lithography shot
- Question : How to overcome scalability limit?**



Road to Future: Resource Disaggregation!

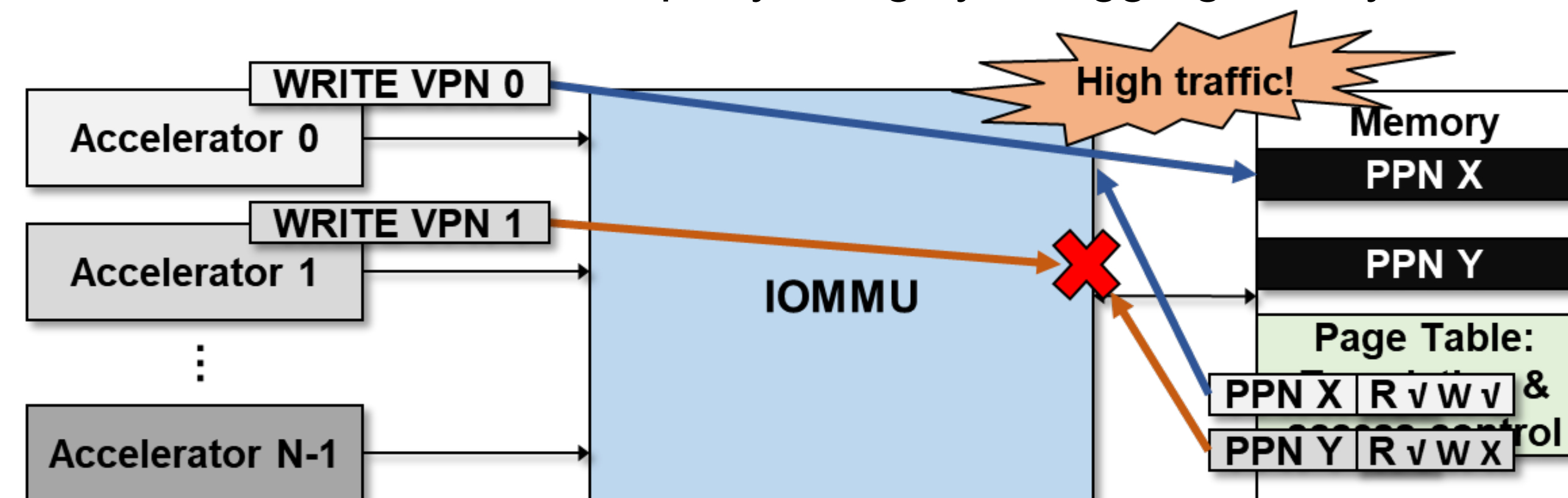
- Alternative architecture? **Resource disaggregation!**
 - Aggregate number of similar components
 - Defined either at chip-level (chiplet) or system-level
 - Goal** Higher scalability, availability, and robustness
- Fundamental problem: Traffic congested on network!**
 - Metadata traffic: security & compression metadata...
 - Data traffic: user- or application-level data



Topic 1: Security—The Red Line of Systems

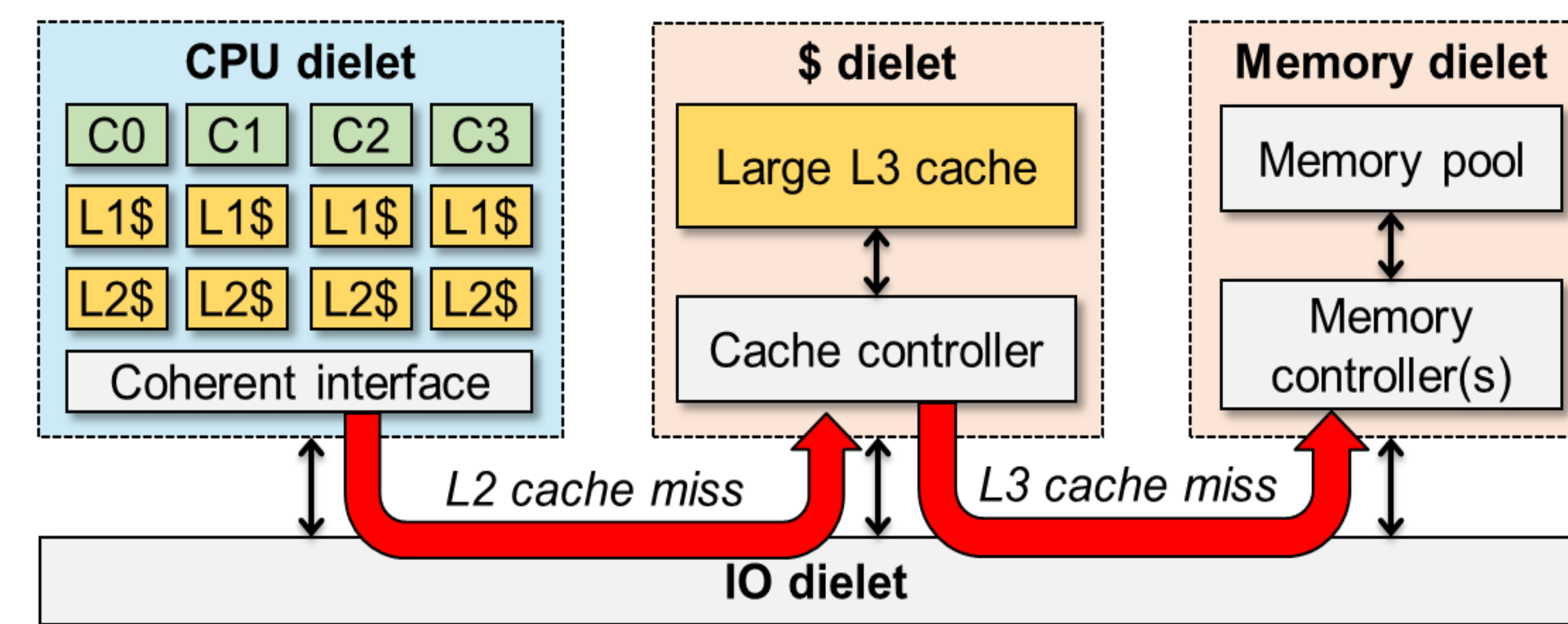
Sandboxing from Malicious Third-Party IPs

- Develop secure architecture while disaggregating accelerators
- Mission 1: Sandbox from malicious accelerators
 - Address translation is common in accelerator-rich systems
 - Accelerators may manipulate translation & permission info.
- Mission 2: Sandbox from malicious IOMMU
 - IOMMU conducts address translation
 - IOMMU can be 3rd party in highly disaggregated systems



Sandboxing from Malicious Memory Controller (MC)

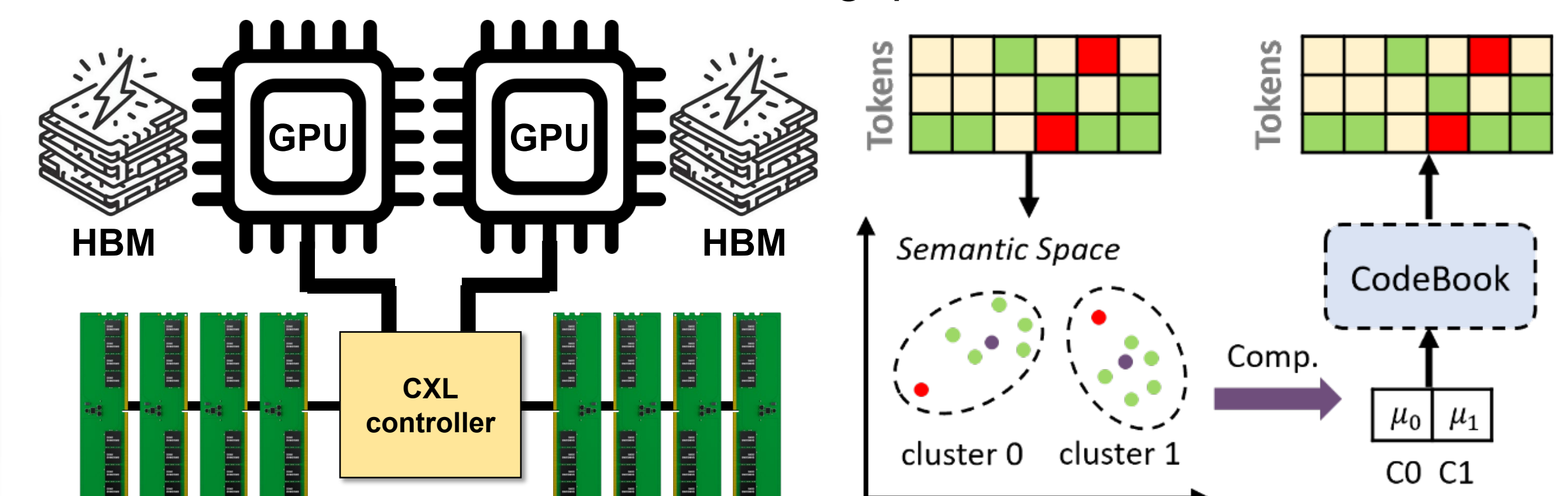
- Why 3rd party MC? What's wrong with it?
 - Chiplet may import MC with complicated policy as 3rd party
 - MC could be tape-out dielet → Memory encryption engine (MEE) now shouldn't be in MC
- Potential architectural issues of decoupling MEE from MC
 - Latency exposure** due to faster data fetch from L3 cache
 - Cache pollution on L3 cache** would be problematic for workloads having huge working set



Topic 2: Compression—Towards High Availability

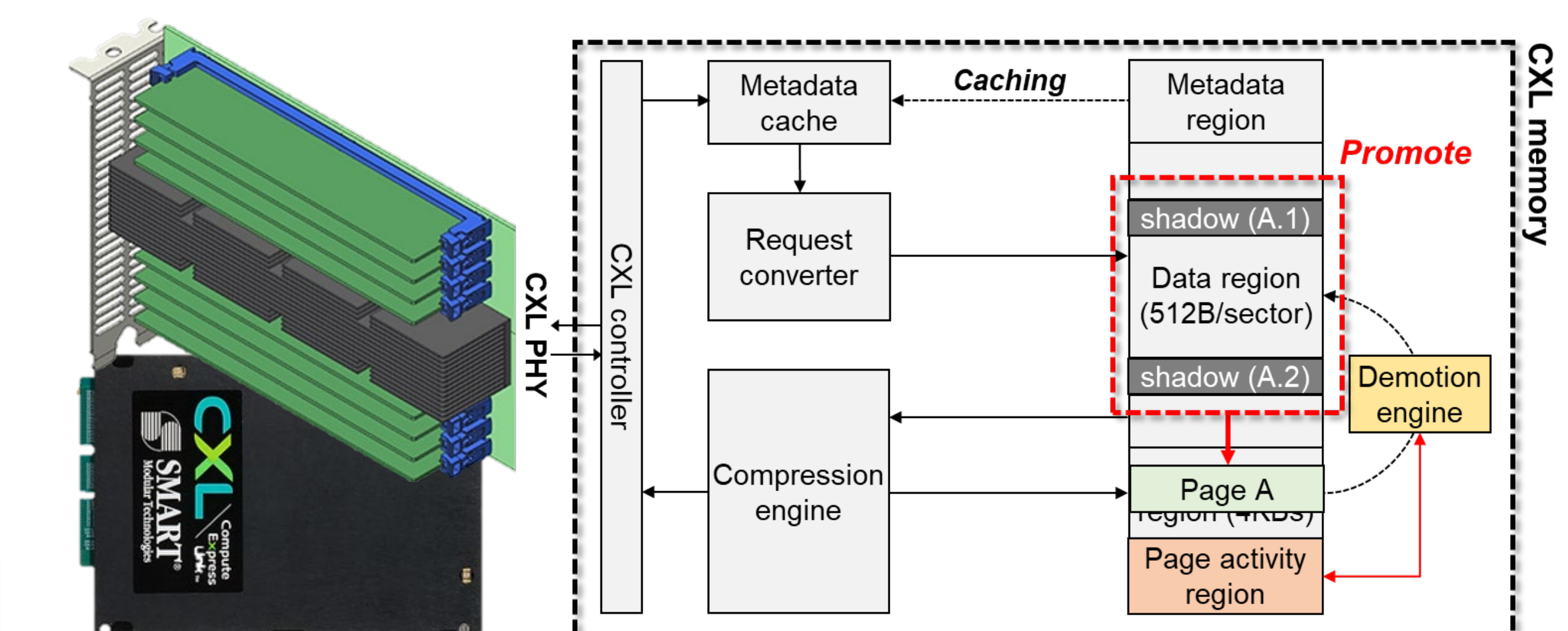
Lossy Compression for Large-Language Model (LLM)

- KV cache of modern LLM inference is a troublesome
 - KV Cache? Context information representing history
 - More tokens, larger KV cache → larger memory is needed
- CXL memory is a key to accommodate large KV cache with high cost-efficiency compared to adding more GPUs/NPUs
- Challenge: CXL memory is much slower than HBMs**
 - Think about their bandwidth gap...



Lossless, Form-Factor-Aware Compression for CXL

- CXL is considered a candidate for memory disaggregation
- BUT its form factor limit entails several **practical challenges**
 - Fewer memory devices than expected → Low capacity & bandwidth
 - Memory compression** should be in CXL memory
- Challenge: Compression, especially block level, incurs bulky data movements within CXL memory**



Group Information & Achievements

- Members Now: 2 Ph.D. candidates 2 Integrated + M.S.-Ph.D. course (석박통합과정) students
- * Recruiting Integrated M.S.-Ph.D. course! 석박통합과정 모집중!**
- Published at 9 top-tier conference venues (MICRO, HPCA, PACT, ICS...), 8 top-tier journals

Benefits

- Close mentorship with tuition fee exemption
- Attending international conferences
- Accessible to valuable infrastructure